TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Analysis of Knit Fabrics: General Procedure, Determination of Specifications for Reproduction, and Production Calculations

The knitted counterpart of Study 1: from an unlabelled knitted sample, produce its full specification — structure, wale and course densities, stitch length, yarn count and weight — and then estimate the machine that made it and the rate at which it can be reproduced. All measurements are worked on one single-jersey sample, following the methods of TE-3102 and the identities of TE-3205 Topics 9–12.

TE-3206 · FABRIC STRUCTURE AND DESIGN SESSIONAL · STUDY 2 · LEVEL 3 TERM 2

1. Object and order of work

Object: to determine, from a knitted sample, the complete specification needed to reproduce it (Section 8), including a reasoned estimate of the producing machine and its output (Section 7). Order of work — non-destructive first, exactly as in TE-3102 Study 5:

1. **Identify the structure** (Section 2) from loop facing, beds and unravelling behaviour.
2. **Count WPC and CPC** on the relaxed sample (Section 3) and state the relaxation state with them.
3. **Measure stitch length** from an unravelled course over 100 wales (Section 4).
4. **Determine the yarn count** from unravelled yarn — length and weight (Section 5).
5. **Measure GSM** by cutter and cross-check it by the identity (Section 6).
6. **Estimate the machine** — gauge, diameter, needles, feeders — and **calculate production** (Section 7).

Condition of the sample. All geometry is measured on fabric **relaxed and flat**, away from edges, folds and faults, and the state (here: dry-relaxed, conditioned on the bench) is recorded with the results — TE-3205 Topic 10's rule that a knitted count without its state is not a measurement.

2. Identifying the structure

- **Find wales and courses:** wales run along the loop columns; on the technical face they show as vertical ribs of V legs; courses run across them.
- **Weft or warp knitted?** Tease an edge yarn: this sample unravels course by course → **weft knitted**.
- **Which structure?** All loops are drawn to the same side — face shows V legs throughout, back shows only head and sinker arcs; edges curl (face-ward top and bottom, back-ward at the sides). No alternating wales (not rib), no second rib behind (not interlock), no course-wise alternation (not purl). **Structure: single jersey (plain weft knit)** — sketch three wales × three courses in the report as the identification record (Figure 1).

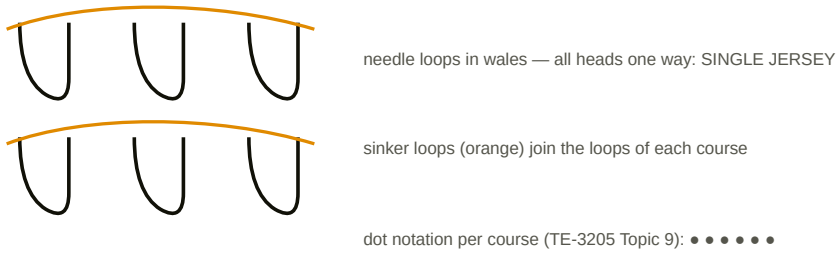


Figure 1 — Loop sketch of the sample's structure on the technical face (the identification sketch your report is marked on): single jersey, every loop drawn the same way on one needle bed.

3. Wales and courses per centimetre

Apparatus: counting glass or lens with a centimetre scale; needle pointer. Count at **three places** on the relaxed sample and average (TE-3102 method).

> **WORKED RESULT — DENSITIES (DRY-RELAXED STATE)**
>
> Wales: 12, 12, 12 per cm → **WPC = 12 per cm**. Courses: 19, 19, 19 per cm → **CPC = 19 per cm**. Stitch density = 12 × 19 = **228 loops per cm²**. In inches: ≈ 30.5 wales and ≈ 48.3 courses per inch.

4. Stitch length by unravelled yarn

Method (TE-3102): mark 100 wales along one course, unravel that course, straighten the yarn **without stretching it**, and measure.

> **WORKED RESULT — STITCH LENGTH**
>
> The course from 100 marked wales measured **300 mm** straightened → stitch length = 300 ÷ 100 = **3.0 mm** (two further courses agreed within 0.03 mm).
>
> Tightness factor: K = √tex ÷ l — with the count of Section 5: K = √19.7 ÷ 3.0 = 4.44 ÷ 3.0 = **1.48**, firm-normal for cotton jersey (TE-3205 Topic 10 scale).

5. Yarn count from unravelled yarn

WORKED RESULT — YARN COUNT

Unravelled yarn: total straightened length **25.0 m**, weight **0.4925 g**.

tex = 0.4925 × 1,000 ÷ 25.0 = **19.7 tex**; Ne = 590.5 ÷ 19.7 ≈ **30 (Ne 30)**.

Report as "19.7 tex (≈ Ne 30), from unravelled yarn" — method stated with the number, as in TE-3102.

6. Fabric weight — measured and calculated

Measured by GSM cutter: the 100 cm² disc weighed **1.36 g** → GSM = 1.36 × 100 = **136 g/m²** (three discs averaged, spread small).

WORKED CROSS-CHECK — GSM BY THE IDENTITY (TE-3205, TOPIC 10)

GSM = WPC × CPC × l × tex ÷ 100 = 12 × 19 × 3.0 × 19.7 ÷ 100 = 228 × 3.0 × 19.7 ÷ 100 = 13,475 ÷ 100 ≈ **135 g/m²**.

Against 136 g/m² measured: agreement within about 1% — accept. (The cutter weighs the fabric as it is, moisture and finish included; the identity counts yarn only. A gap beyond a few per cent sends you back to the stitch-length measurement first — stretched yarn is the usual culprit.)

7. Machine estimate and production calculation

The sample's geometry constrains the machine that can have made it:

- **Gauge.** Relaxed wales per inch = 12 × 2.54 ≈ 30.5. On the machine the fabric is held wider than its relaxed state, so the needle spacing must be *coarser* than 30.5 per inch: gauge 28 gives 28 needles/inch, i.e. the fabric has contracted in width by (30.5 − 28) ÷ 30.5 ≈ 8% on relaxation — the normal direction and size of change for cotton jersey, and the standard gauge for Ne 30 jersey. **Estimate: 28 gauge.** (Gauge 24 would demand ~21% width contraction — implausible; gauge is estimated by this consistency argument, not guessed.)
- **Diameter and needles.** For a garment-length fabric of about 139 cm relaxed open width (Topic 10's relation: width = needles ÷ WPC), choose a **19-inch diameter** cylinder: needles = π × 19 × 28 = **1,671 needles**; tube circumference on the machine = π × 19 in ≈ 152 cm, and 1,671 ÷ 12 ≈ **139 cm** relaxed open width — consistent both ways.
- **Feeders and speed.** A typical single-jersey machine of this diameter carries about 3 feeders per inch of diameter → **56 feeders**; run at **24 rpm**.

WORKED PRODUCTION CALCULATION (VERIFY EACH LINE)

Courses per minute = feeders × rpm = 56 × 24 = **1,344 courses/min**.

Yarn per course = needles × stitch length = 1,671 × 3.0 mm = 5,013 mm ≈ **5.01 m per course**.

Yarn consumption = 1,344 × 5.01 = 6,737 m/min × 19.7 tex ÷ 1,000 = 132.7 g/min ≈ **7.97 kg/h at 100% efficiency**; at 85% running efficiency ≈ **6.77 kg/h**.

Length output (relaxed fabric, 136 g/m², 1.39 m width): area rate = 6,770 ÷ 136 ≈ 49.8 m²/h → length = 49.8 ÷ 1.39 ≈ **35.7 m of fabric per hour**.

Note the method: production is calculated from **yarn mass balance** (courses × yarn per course), never from CPC alone — the on-machine course density differs from the relaxed CPC of Section 3, and only the mass chain is state-independent.

8. The specification sheet — the finished result

ITEM	RESULT FOR THE SAMPLE OF THIS SHEET
Structure	Weft-knitted single jersey (plain); face/back identified, sketch Figure 1
WPC / CPC (dry-relaxed)	12 per cm / 19 per cm (averages of three counts); stitch density 228 per cm²
Stitch length	3.0 mm (unravelled course, 100 wales, 300 mm); tightness factor K = 1.48
Yarn count	19.7 tex ≈ Ne 30 (25.0 m unravelled yarn, 0.4925 g)
Fabric weight	136 g/m² by cutter (1.36 g disc); 135 g/m² calculated — agreement within ~1%
Fibre	By burning test and microscopy: cotton (state your own observations for your sample)
Machine estimate	Circular single-bed, 28 gauge, 19-inch diameter, 1,671 needles, 56 feeders (estimate by the contraction argument of Section 7)
Reproduction rate	At 24 rpm: 7.97 kg/h at 100%, ≈6.77 kg/h at 85% efficiency; ≈35.7 m/h of relaxed fabric at 139 cm width

9. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. An unravelled course from 100 wales measures 300 mm, and 25.0 m of unravelled yarn weighs 0.4925 g. Give the stitch length and the count in tex and Ne.

2. From WPC 12, CPC 19, l = 3.0 mm and 19.7 tex, calculate the expected GSM and comment on a cutter result of 118 g/m².
3. Why is the producing machine's gauge (28) *coarser* than the relaxed fabric's wales per inch (30.5)? What assumption would a gauge-32 estimate require?
4. Why is production calculated from yarn mass balance rather than from courses per minute and relaxed CPC?

Answer key

1. Stitch length = 300 ÷ 100 = **3.0 mm**. tex = 0.4925 × 1,000 ÷ 25.0 = **19.7 tex**; Ne = 590.5 ÷ 19.7 ≈ **30**.
2. Expected GSM = 12 × 19 × 3.0 × 19.7 ÷ 100 ≈ **135 g/m²**. A cutter result of 118 is some 13% low — well outside agreement, so something measured is wrong. Suspect a **state mismatch** first: densities counted on a more-relaxed (more contracted) piece than the disc was cut from inflate the calculated value, while a disc cut from fabric held under tension contains less yarn and weighs light. Re-do all three measurements on the same conditioned, relaxed piece; then re-check the stitch length for stretched yarn, and the weighing.
3. Because the fabric contracts in width when it relaxes off the machine: the same needles produce more wales per inch in the relaxed fabric than the needle spacing per inch. A 32-gauge estimate would require the relaxed wales per inch (30.5) to be *less* than the needle density — i.e. the fabric to have widened after leaving the machine — the wrong direction for cotton jersey relaxation.
4. Because courses per minute is fixed by the machine, but the *length* those courses occupy changes with relaxation state: on-machine CPC is lower than relaxed CPC. Yarn mass, by contrast, is conserved — needles × stitch length per course × courses × tex gives the true consumption in any state, and length output follows by dividing by the relaxed GSM and width.
